RAJALAKSHMI ENGINEERING COLLEGE

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A2

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|---|---|
| **Started on** | Tuesday, 22 September 2026, 9:07 AM |
| **State** | Finished |
| **Completed on** | Tuesday, 22 September 2026, 9:08 AM |
| **Time taken** | 1 min 6 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

**For example:**

| Input | Result |
|---|---|
| 5<br>1 1 2 3 4 | 1 |

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>

int findDuplicate(int arr[], int n) {
    int slow = arr[0];
    int fast = arr[0];

    do {
        slow = arr[slow];
        fast = arr[arr[fast]];
    } while (slow != fast);

    slow = arr[0];

    while (slow != fast) {
        slow = arr[slow];
        fast = arr[fast];
    }

    return slow;
}

int main() {
    int n;
    scanf("%d", &n);

    int arr[n];

    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    printf("%d", findDuplicate(arr, n));

    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 11<br>10 9 7 6 5 1 2 3 8 4 7 | 7 | 7 | ✔ |
| ✔ | 5<br>1 2 3 4 4 | 4 | 4 | ✔ |
| ✔ | 5<br>1 1 2 3 4 | 1 | 1 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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